

Nishant Joshi

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Research Software Engineer with a PhD in Computational Neuroscience and 6+ years developing reproducible Python workflows, open-source analysis tools, and HPC-enabled pipelines for large-scale biological data. Experience working at the intersection of academic research and software engineering, with strengths in scientific Python, Docker, version control, FAIR data practices, and collaborative method development. Motivated to build sustainable research software that enables scientists across disciplines to answer complex questions.

PROFESSIONAL EXPERIENCE

PhD researcher | Donders Institute | Radboud University – Nijmegen, NL

May 2021 – Dec 2025

- Built and operated end-to-end analysis pipelines for large-scale single-cell electrophysiology (time-series) datasets.
- Processed 3.6B data points from **high-throughput biological readouts**, building robust ingestion, validation, and high dimensional feature extraction workflows, handling real-world data challenges like missing values, misalignments, noise and formatting.
- Conducted multivariate and clustering analyses (Louvain, Leiden algorithms) on high-dimensional neural population datasets, communicating results to cross-functional research teams.
- Applied multi-set correlation and factor analysis (MCFA) to integrate four high-dimensional feature sets across single-cell recordings, identifying which biological attributes best explain functional heterogeneity across neuronal populations.
- Packaged models and pipelines using Docker with version control and reproducible releases, enabling reuse and deployment across multiple research teams and external institutions (Netherlands eScience Centre, U-Ghent).
- Mentored 4 students (2 BSc, 2 MSc) in Python programming, version control, HPC workflows, and reproducible data management.

Visiting Data Scientist | Netherlands eScience Centre – Amsterdam, NL

Feb 2023 – May 2023

- Designed and enforced FAIR data principles (metadata, versioning, interoperability) across analysis workflows, enabling researchers to reliably reuse code and data.
- Delivered maintainable opensource python based ML toolkit for extracting high-dimensional electrophysiological features from noisy timeseries data.
- Spearheaded a 2 year strong collaboration, resulting in 2 publications.

Visiting Researcher | IMBB-FORTH – Iraklion, GR

Feb 2024 – Apr 2024

- Developed computational pipelines to fit biophysical models to single-cell recordings, extracting and interpreting quantitative biological parameters to characterise functional cell-type diversity.

ML Engineer | INM 6, Forschungszentrum Jülich – Jülich, DE

Feb 2020 – Apr 2021

- Designed, trained and evaluated 100+ RNN and SNN architectures using distributed GPU training (Horovod) on JURECA HPC, studying dynamical scaffolding (fixed point analysis) and representational similarity (SVCCA).
- Achieved **24x reduction in model training time** by optimizing distributed computing workflows on HPC infrastructure.
- Owned the full ML lifecycle, from preprocessing and training to automated evaluation, supporting downstream decision-making and scientific inference.

Project Intern | Qure.ai – Mumbai, IN

Jun 2017 – Sep 2017

- Engineered automated image segmentation and annotation pipelines for medical CT scans, training neural networks to detect brain haemorrhages.
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EDUCATION

PhD – Computational Neuroscience (thesis accepted)

May 2021 – Dec 2025

Radboud University | Donders Institute for Brain, Cognition and Behaviour, Nijmegen, NL

Dual Degree MSc – Computer Science and Engineering

Oct 2018 – Nov 2020

Grade: 9.1/10, Thesis grade: 10/10

KTH Royal Institute of Technology – Stockholm, SE and TU Berlin – Berlin, DE

Bachelor of Engineering – Mechanical/Industrial Engineering

Aug 2013 – May 2017

Grade: First Class with distinction, Thesis grade: 10/10

R.V. College of Engineering – Bangalore, India

SKILLS

- **Data & Software:** Python, R, SQL, Bash, Unix, Git, Docker, CI/CD, MLflow, FastAPI, C/C++
 - **Scientific Computing:** Distributed computing, HPC workflows, workflow reproducibility, FAIR data practices
 - **ML & Analysis:** PyTorch, TensorFlow, scikit-learn, multivariate analysis, clustering, time-series modelling
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ACHIEVEMENTS

- Marie Skłodowska-Curie Actions PhD Fellowship (Flagship EU PhD fellowship)
 - EIT Digital Excellence Scholarship (Top 5%)
 - First author publication in PLoS Computational Biology
 - Best Bachelor's Thesis Award
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PUBLICATIONS

1. *Joshi N et al. (2025) Neuronal identity is not static. PLoS Comput Biol 21(12): e1013821.*
2. *Joshi N et al. Neuromodulatory Control of Cortical Function. bioRxiv 2026.03.13.711516. (Under revision at PNAS)*
3. *Iacob† S, Joshi† N et al. Timescale Heterogeneity in Reservoir Computing. In preparation. (†co-first author)*